NSE Cross-Sectional Alpha Research: do classic price signals survive Indian costs and an honest holdout?

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Abstract

We ask whether eight classic daily price and volume signals predict cross-sectional returns on the National Stock Exchange of India after realistic costs, using a survivorship-free point-in-time panel built from exchange archives, a pre-registered protocol, a trial ledger and a locked out-of-sample window read once. Development (walk-forward, 2017–2024): long-only top-decile book net information ratio not run (gross not run), deflated Sharpe not run with not run logged trials. Holdout (2024-11-01 to 2026-10-30): net IR not run, 95% bootstrap interval [not run, not run]; F&O long-short net Sharpe not run. Every number in this paper is rendered from the run files by `qr ledger`.

1 Introduction

Do classic price signals survive Indian transaction costs and an honest holdout? Three things make this answer checkable. First, a survivorship-free daily NSE panel (88 sessions, 2724 securities, 0 later-delisted names that were in the liquid universe) rebuilt from raw exchange files. Second, a locked protocol: a pre-registration tag before the first IC, an append-only trial ledger, and a holdout read exactly once. Third, a cost model that changes with the tax rules, including the April 2026 rise in stock-futures STT from 0.02% to 0.05%, which falls inside the holdout window.

2 Data

Raw files come from the NSE static archives (cash bhavcopy in the legacy and UDiFF formats, delivery files, F&O bhavcopy, index files), downloaded at no more than one request per second with an honest User-Agent and an on-disk cache. A permanent security id links symbol changes through ISIN. Prices are adjusted to total returns for splits, bonuses and dividends from NSE corporate-action exports; rights issues are not adjusted, demergers are nulled, and any residual one-day move above 25% is classified by a jump audit and nulled if unexplained. The universe is rebuilt each month from data dated before the month starts: EQ/BE series, at least 252 sessions of history, at least 55 trades in 63 sessions, close at least Rs 10, median traded value at least Rs 1 crore, top 750 by value. The Nifty 500 total-return index is approximate (price index plus dividend yield) unless an official file is supplied. Data-quality gates and the survivorship counts are in the ledger (`data.*`).

3 Method

3.1 Signals and evaluation

Eight signals with signs and windows fixed before any IC was computed (`PREREG.md`): MOM_12_1 [Jegadeesh and Titman, 1993], REV_5, LOWVOL_63, IVOL_63 [Ang et al., 2006], AMIHUD_63 [Amihud, 2002], ABVOL_5_63, DELIV_ABN and HIGH52 [George and Hwang, 2004]. Each is winsorized at 1/99, standardized by date, neutralized to sector dummies, a shrunk Dimson beta and size, and re-standardized. We report rank IC at horizons 1, 5, 10 and 21 sessions with Newey–West t statistics [Newey and West, 1987] and Holm adjustment across signals; a signal passes at $t \geq 3$ after Holm correction.

3.2 Combination

Equal weight, ridge and LightGBM are compared on eight expanding-window folds with a 27-session purge and embargo. The headline is the better of ridge and LightGBM only if it beats equal weight with Newey–West $t \geq 2$ on the IC difference, otherwise equal weight [López de Prado, 2018, Gu et al., 2020].

3.3 Portfolios and costs

P1 is a long-only equal-weight book (enter in the top 10%, hold to the top 20%, weekly, signal at the close and fill at the next close, 10% of ADV cap) against Nifty 500 TRI. P2 is long the top quintile in cash and short the bottom quintile in near-month stock futures. Costs are per side: dated STT, exchange charge, SEBI fee, stamp duty, brokerage (assumed 3 bp delivery, 1 bp futures), GST and the DP charge, plus an Abdi–Ranaldo half-spread [Abdi and Ranaldo, 2017] and square-root impact $Y\sigma\sqrt{Q/\mathrm{ADV}}$ with $Y = 1$. Capacity is the smaller of the AUM at which the net metric halves and the AUM at which more than 20% of desired trading is blocked by the ADV cap.

3.4 Validation protocol

A pre-registration commit and tag precede the first IC; `oos.lock` makes every loader blind after 2024-10-31; every signal, combiner fold and base portfolio is logged to the trial ledger before results print; the deflated Sharpe ratio [Bailey and López de Prado, 2014] uses the true ledger count; confidence intervals use a stationary block bootstrap [Politis and Romano, 1994].

4 Results

4.1 Single signals

Strongest neutralized signal: not run, mean rank IC not run (t_{NW} = not run, $h = 5$). Signal tests passing Holm and $t \geq 3$: not run. figure not generated (run `qr figures`)

4.2 Combination

Headline combiner: not run (rank IC not run; equal weight not run; paired difference t = not run).

	P1 gross	P1 net	P2 gross	P2 net
Information ratio / Sharpe	not run	not run	not run	not run
CAGR (net)		not run		not run
Max drawdown (net)		not run		not run
One-way turnover per year		not run		not run
Cost drag, bp per year		not run		not run

Table 1: Development window 2017-01-02 to 2024-10-31 (walk-forward). Risk-free: constant 6.5% fallback.

	P1 gross	P1 net	P2 gross	P2 net
Information ratio / Sharpe	not run	not run	not run	not run
95% CI (net)		[not run, not run]		[not run, not run]
PSR(0)		not run		not run
Cost drag, bp per year		not run		not run
STT share of drag, %		not run		not run

Table 2: Locked out-of-sample window 2024-11-01 to 2026-10-30, read once.

4.3 Development portfolios

4.4 The holdout read

figure not generated (run `qr figures`)

4.5 Capacity

P1 capacity: Rs not run crore; break-even cost multiple not run. figure not generated (run `qr figures`)

5 Robustness

Cost sensitivities (brokerage 0 and 5 bp, impact $Y \in \{0.5, 2\}$, costs doubled) and a 30% delisting haircut are written to `sensitivities_p*.csv` in the portfolio run; they are not new alpha trials and are not in the ledger. Further checks listed in the PRD (no sector neutralization, monthly rebalance, by-year results) are reported only if run.

6 Limitations

- Sector labels are a 2026 snapshot, not point-in-time; names delisted earlier are `Unclassified`.
- No shares-outstanding data: size is proxied by traded value; share turnover is not tested.
- Corporate-action coverage is only as good as NSE's export; unresolved jumps are nulled (counts in the data-quality report).
- True delisting returns are unknown; a held name carries flat for up to 10 sessions and is then written off at a 0% or 30% haircut.
- Statutory rates were spot-checked from public sources; the 2024-10-01 futures STT date and the pre-2016 delivery history are not independently confirmed (no reported number depends

on pre-2016 rates). Brokerage is assumed; futures margin is a flat 20%; lot rounding is ignored.

- The impact model is a square-root law with $Y = 1$, not fitted to Indian execution data. P2 uses cash-market ADV and volatility for the futures leg.
- The Nifty 500 total-return series is approximate unless an official file was supplied; the risk-free rate is a constant 6.5% fallback.
- The trial count is a lower bound: choosing these eight signals from the literature was itself a selection the ledger cannot see. The paper's own deflated-Sharpe worked example was not reproduced (the PSR example was).
- The researcher lived through 2024–2026 markets; the lock prevents numeric peeking, not general knowledge.
- One market, about eight years of walk-forward plus two of holdout.

7 Conclusion

The conclusions are drawn from the ledger values above, whatever they are: a holdout interval that includes zero or a deflated Sharpe below 0.95 is reported as a null result, with the cost attribution.

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